

Exercise sheet 7

This assignment is due on March 13th at noon, to be submitted electronically via Blackboard.

1. Let $H_\ell(x)$, $\ell \in \mathbb{Z}_{\geq 0}$ be the classical Hermite polynomial defined via the contour integral

$$H_\ell(x) := \frac{\ell!}{2\pi i} \int_{\Gamma} e^{2xz - z^2} \frac{dz}{z^{\ell+1}}, \quad (1)$$

where $\Gamma = \{z \in \mathbb{C} : |z| = 1\}$ is the unit circle in the complex plane traversed once counterclockwise.

- (a) Find $H_0(x)$, $H_1(x)$, and $H_2(x)$ the first three Hermite polynomials.
- (b) Confirm that these first three polynomials are mutually orthogonal with respect to $\exp(-x^2)$.
- (c) Hence find $K_3(x, y)$ for the (renormalised) GUE.

2. Using the differential equation

$$\frac{d^2}{dx^2} H_\ell(x) - 2x \frac{d}{dx} H_\ell(x) + 2\ell H_\ell(x) = 0$$

for the Hermite polynomials (1), show that $\psi_\ell^{(N)}(x) = e^{-\frac{N}{2}x^2} N^{\frac{1}{4}} \gamma_\ell H_\ell(\sqrt{N}x)$ with $\gamma_\ell = 1/\sqrt{\sqrt{\pi}2^j \cdot j!}$, satisfies the second order equation

$$\frac{d^2}{dx^2} \psi_\ell^{(N)}(x) = N(Nx^2 - (2\ell + 1)) \psi_\ell^{(N)}(x).$$

3. Define

$$\psi_j(x) = N^{1/4} \cdot \gamma_j \cdot \exp\left(-\frac{N}{2}x^2\right) \cdot H_j(\sqrt{N}x)$$

for $\gamma_j = 1/\sqrt{\sqrt{\pi}2^j \cdot j!}$ (so $\psi_j(x) = \sqrt{w_N(x)} p_j(x)$ is the ‘rescaled’ orthogonal polynomials for the GUE after rescaling by $1/\sqrt{N}$). This (tricky) exercise is to prove (16.13) in the notes: Using the Plancherel-Rotach asymptotics (cf. Proposition 16.1, expression (16.8)) show that as $N \rightarrow \infty$,

$$\psi_{N-k}(x) = \sqrt{\frac{2}{\pi\sqrt{2-x^2}}} \cos\left(N\beta(\theta) - \left(k - \frac{1}{2}\right)\theta - \frac{\pi}{4}\right) + \mathcal{O}\left(\frac{1}{N}\right).$$

Here, $\beta(\theta) := \theta - \frac{1}{2} \sin(2\theta)$ and $x = \sqrt{2} \cos \theta$, $\theta \in (0, \pi)$ lies inside $(-\sqrt{2}, \sqrt{2})$.